SIMATS ENGINEERING
ASSESSMENT TOOL 2: Data Interpretation
COURSE NAME: Fundamentals of Data Science COURSE CODE: DSA0405
Slot D

COURSE OUTCOME COVERED

CO3: Analyze datasets using descriptive statistical measures such as mean, variance, sampling methods, covariance, and correlation to identify patterns, relationships, and support data-driven decision-making.

Assessment Tool Weightage: 40%

QUESTIONS:5 **Total Marks 50** **Duration :60 Minutes**

Annexure A
Data Interpretation

Code of Conduct:
I Y. Jayesh Ranjan- 192472201 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: Y. Jayesh Ranjan

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	*Marks(100)*
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

SIMATS ENGINEERING

COURSE NAME: FUNDAMENTALS OF DATA SCIENCE

COURSE CODE: DSA0404 & DSA0405

CO 3 - ASSESSMENT TOOL 2: Data Interpretation

COURSE OUTCOME COVERED

- Apply data preparation and exploratory data analysis techniques, including data summarization, data distribution analysis, outlier treatment, and measures of asymmetry, to preprocess and interpret datasets effectively.
- Analyze datasets using descriptive statistical measures such as mean, variance, sampling methods, covariance, and correlation to identify patterns, relationships, and support data-driven decision-making.

1. Standardization/Normalization
A feature "Age" ranges from 18–75 and another feature "Income" ranges from ₹20,000–₹20,00,000 in a dataset meant for a distance-based algorithm. Explain, with reasoning (not just definitions), why failing to standardize these variables would distort analysis, and calculate the z-score for Age = 45 if mean = 40 and SD = 12.

2. Distribution Shape from Summary Stats Alone
A dataset has Mean = 45, Median = 45, Mode = 45, and skewness ≈ 0, but kurtosis = 5.8 (leptokurtic). Explain what this combination tells you about the distribution's shape that skewness and central tendency alone would miss.

3. Grouped Data — Mean and Variance
Class Interval: 0–10, 10–20, 20–30, 30–40, 40–50
Frequency: 5, 8, 15, 12, 10
Calculate the mean and variance using the step-deviation method. Show the assumed midpoints and working.

4. Comparing Two Distributions via CV
Portfolio A: Mean return = 12%, SD = 3%
Portfolio B: Mean return = 18%, SD = 5%
An investor claims Portfolio B is "riskier in absolute terms but not necessarily worse per unit of return." Evaluate this claim quantitatively using Coefficient of Variation and explain what CV captures that SD alone doesn't.

5. Outlier's Effect on Variance
A dataset of 10 values has variance = 25. One value, originally 40 (close to the mean of 42), is mistakenly recorded as 400. Without recalculating exactly, explain directionally and reason about the magnitude of distortion to (a) the mean, (b) the variance — and why variance is affected disproportionately more than the mean.

Answers:

1. Standardization / Normalization

Given:

$x = 45$

Mean, $\mu = 40$

SD, $\sigma = 12$

Z-score formula

$z = (x - \mu) / \sigma$

x = observed value; μ = mean; σ = standard deviation.

Substitute:

$$z = (45 - 40) / 12$$

$$z = 5 / 12$$

$$z = 0.4167$$

Therefore,

$$\mathbf{z = 0.417}$$

Interpretation: Age 45 is 0.417 standard deviations above the mean.

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2. Distribution Shape

Given:

Mean = 45

Median = 45

Mode = 45

Skewness ≈ 0

Kurtosis = 5.8

Since:

Mean = Median = Mode = 45

the distribution is centered at 45.

Skewness ≈ 0 indicates approximately symmetric distribution.

For kurtosis:

Kurtosis > 3 → Leptokurtic

Therefore:

5.8 > 3

Hence, the distribution is **leptokurtic**.

Final interpretation:

Approximately symmetric and leptokurtic, with heavier tails.

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3. Grouped Data – Mean and Variance

Given:

Class	f
0–10	5
10–20	8
20–30	15
30–40	12
40–50	10

Step 1: Midpoints

Formula:

$$x = (\text{Lower limit} + \text{Upper limit}) / 2$$

x = class midpoint.

0–10:

$$x = (0 + 10) / 2 = 5$$

10–20:

$$x = (10 + 20) / 2 = 15$$

20–30:

$$x = (20 + 30) / 2 = 25$$

30–40:

$$x = (30 + 40) / 2 = 35$$

$$40–50:$$

$$x = (40 + 50) / 2 = 45$$

Take:

$$A = 25$$

$$h = 10$$

Step 2: Step deviations

Formula:

$$d' = (x - A) / h$$

d' = step deviation; x = midpoint; A = assumed mean; h = class width.

For x = 5:

$$d' = (5 - 25) / 10 = -2$$

For x = 15:

$$d' = (15 - 25) / 10 = -1$$

For x = 25:

$$d' = (25 - 25) / 10 = 0$$

For x = 35:

$$d' = (35 - 25) / 10 = 1$$

For x = 45:

$$d' = (45 - 25) / 10 = 2$$

Step 3: Working table

Class	f	x	d'	fd'	fd'²
0–10	5	5	-2	-10	20
10–20	8	15	-1	-8	8
20–30	15	25	0	0	0
30–40	12	35	1	12	12
40–50	10	45	2	20	40
Total	**50**			**14**	**80**

Therefore:

$$\Sigma f = 50$$

$$\Sigma fd' = 14$$

$$\Sigma fd'^2 = 80$$

Step 4: Mean

Formula:

$$\text{Mean} = A + h(\Sigma fd' / \Sigma f)$$

A = assumed mean; h = class width; $\Sigma fd'$ = sum of fd'; Σf = total frequency.

Substitute:

$$\text{Mean} = 25 + 10(14 / 50)$$

$$= 25 + 10(0.28)$$

$$= 25 + 2.8$$

Mean = 27.8

Step 5: Variance

Formula:

$$\text{Variance} = h^2[\Sigma fd'^2 / N - (\Sigma fd' / N)^2]$$

h = class width; N = total frequency; $\Sigma fd'^2$ = sum of fd'^2; $\Sigma fd'$ = sum of fd'.

Substitute:

$$\text{Variance} = 10^2[80 / 50 - (14 / 50)^2]$$

$$= 100[1.6 - (0.28)^2]$$

$$= 100[1.6 - 0.0784]$$

$$= 100(1.5216)$$

Variance = 152.16

Final

$$\textbf{Mean = 27.8}$$

$$\textbf{Variance = 152.16}$$

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4. Comparing Two Distributions Using CV

Given:

Portfolio A:

$$\text{Mean} = 12\%$$
$$\text{SD} = 3\%$$

Portfolio B:

$$\text{Mean} = 18\%$$
$$\text{SD} = 5\%$$

Step 1: CV of Portfolio A

Formula:

CV = (SD / Mean) × 100

CV = relative variability; SD = standard deviation; Mean = mean return.

CV(A) = (3 / 12) × 100

= 0.25 × 100

CV(A) = 25%

Step 2: CV of Portfolio B

CV(B) = (5 / 18) × 100

= 0.277777... × 100

CV(B) = 27.78%

Step 3: Comparison

CV(A) = 25%

CV(B) = 27.78%

Since:

27.78% > 25%

Portfolio B has greater relative variability.

Also:

SD(B) = 5% > SD(A) = 3%

Therefore, Portfolio B has greater absolute variability as well.

Final

Portfolio A: CV = 25%

Portfolio B: CV = 27.78%

Portfolio B is riskier both in absolute terms and relative to its return.

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5. Outlier's Effect on Variance

Given:

n = 10

Original value = 40

Mean = 42

Incorrect value = 400

Original variance = 25

Step 1: Effect on Mean

Formula:

Mean = Σx / n

Σx = total of observations; n = number of observations.

Change in the value:

400 - 40 = 360

Therefore, total sum increases by:

360

Since $n = 10$:

Change in mean = 360 / 10

= 36

So the mean moves upward substantially.

Using the original mean only to show the direction:

New mean ≈ 42 + 36

≈ 78

Therefore:

Mean increases substantially.

Step 2: Effect on Variance

Formula:

$$\text{Variance} = \Sigma(x - \text{Mean})^2 / n$$

x = observation; Mean = average; n = number of observations.

For the original value:

$x = 40$

Mean = 42

Deviation:

$40 - 42 = -2$

Squared deviation:

$(-2)^2 = 4$

For the incorrect value:

$x = 400$

Using the original mean for comparison:

$400 - 42 = 358$

Squared deviation:

358^2

$= 358 \times 358$

$= 128164$

Comparison:

Original squared deviation = 4

Incorrect squared deviation = 128164

Therefore, the contribution to the squared-deviation total becomes enormously larger.

Hence:

Variance increases drastically.

The reason is directly visible from the calculation:

$2^2 = 4$

whereas

$358^2 = 128164$

Therefore, the outlier has a much greater effect on variance than on the mean.

Final

Mean → increases substantially

Variance → increases drastically

Reason → variance uses squared deviations.